

not only have worked during his college assignments but even otherwise have practiced enough to demonstrate that he is interested in this domain. Assembly, troubleshooting and testing are some of the widely sought skills for this domain."

A good understanding of basic electronics is required. Mitra says, "It's not just the engineering degree, a potential candidate needs to understand how a design looks like and how to interpret the design so that during manufacturing, in case there are some issues, these can be reported. That's the kind of value add that people look for in profiles of the engineers." He adds, "There are also multiple industry certifications that recruiters tend to look at. There are some certifications that engineers might have acquired in the past which could help them stand out among other candidates."

From a product-development perspective, physics and engineering tool knowledge are extremely important, informs Nande. He says, "If you are on the manufacturing side, obviously process and supply-chain knowledge is useful. Increasingly, soft skills and good communication skills are also seen as differentiators in promoting engineers to managerial cadre."

## Demand areas

According to the data provided by TimesJobs.com, the top industries hiring engineers are in the manufacturing and engineering sector with about 28 per cent share of jobs, followed by the IT-software/software services sector with a share of 26 per cent. IT-hardware/semiconductor and healthcare/biotechnology sectors provide 15 per cent and 12 per cent of the jobs, respectively.

Apart from Bengaluru that features in the list of top locations for jobs in electronics manufacturing and SMT sector with 45 per cent share, figures indicate Pune, Delhi and Chennai provide 12 per cent, 12 per cent and 10 per cent of the jobs, respectively. Remaining jobs are scattered in other locations.

## Pay package

This industry will pay well, but only if you have patience. As in any other manufacturing industry, here too, you must be ready to toil, informs Bhargava. Creation of product

## Are certifications worth it?

Yes, certifications are worth it, but mostly at a slightly later stage of career. Normally, these certifications are quite expensive, and though most students acquire these as a part of their engineering curriculum, usually, it is not a viable proposition, according to Mitra. He says, "If somebody is studying and thinking of doing a certification, it is very difficult for recruiters to interpret what the certification is intended to do. Normally, what is preferable is to gather a solid experience of about one or two years and then only go for certification courses. These certification courses cost a significant amount of money in India. Also, it would be difficult for students to interpret, along with their studies, what the trainer is trying to teach."

does not happen without hard work, and that satisfaction is the ultimate one. Talking about his organisation, he says, "Freshers start at about 120,000 rupees per annum, and an experience of a couple of years can fetch them anything between 180,000 and 300,000 rupees, depending on the field and competence of the person." Some reputed organisations offer as much as 220,000 rupees per annum to a fresh trainee engineer.

According to the data provided by TimesJobs.com, 50 per cent of the jobs are for entry-level engineers in the manufacturing domain, and they are paid anywhere between 150,000 and 250,000 rupees, depending on their skills and capabilities. Once they gain experience and elevate to junior-engineer level, their package can become anywhere between 350,000 and 700,000 rupees. Currently, about 26 per cent of the jobs are for fairly experienced, junior-level professionals. About 10 per cent of the jobs are for mid-level and senior-level experts each in the manufacturing domain.

## Expert advice

Anand believes practise is the key. He says, "If you rely only on college/institutional knowledge and practicals, this field may not be a good option for you. People who are interested in making career in SMT and electronics manufacturing industry must show extra-curricular efforts (in these areas) to demonstrate their worth. Doing this ensures their faster progress and growth in industry, once they join."

Interested candidates need to be aligned to the requirements of the firms in India. Students should get more involved and try to identify the industry sector they wish to join, suggests Mitra. Giving an example of SMTA student chapter, he says,

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"SMTA's student chapters are currently located outside India. Within South East Asia, like Malaysia and Thailand, students take advantage of the local chapters to establish relation with the industry." Talking about India, he adds, "We are trying to establish student chapters in cooperation with some institutes, specifically in the southern part of the country. If students are interested, they can approach the SMTA contacts, and we can acquaint them with the industry requirements."

## By 2020 jobs will hit mainstream

At present 75 per cent to 80 per cent of the electronics is imported and only about 20 per cent is manufactured in India, to fulfill its internal demand. So, obviously, there is a huge scope to manufacture within India. Mitra says, "This is the reason why all the top multinational electronics design or manufacturing companies have their offices in India already since the last decade or so. They are just waiting for the regulatory clearances and then they will expand in India the way they had initially planned to." With this being the situation, it definitely makes sense that people align themselves to this industry. Though the industry is picking up, it has not yet moved into that level of maturity. There will be tremendous job opportunities in the coming years, probably by 2020, besides those already existing today. ●

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